PUBLIKASI PROGRAM KONSERVASI KEANEKARAGAMAN HAYATI PT ARUTMIN INDONESIA NPLCT

No.	Program	Media Publikasi	Peruntukan	Tautan
1	Pemetaan Kategori Subtrat Dengan Metode OBIA Menggunakan Citra SPOT-7 Pada Terumbu Karang Pinggiran Di Tanjung Pemancingan, Kabupaten Kotabaru	1. *IOSR Journal of Environmental Science, Toxicology and Food Technology* (IOSR-JESTFT) 2. Arutmin NPLCT CSR Enviro	1. Akademisi 2. LSM 3. Pemerhati Lingkungan 4. Publikasi ke Publik	Klik disini untuk IOSR-JESTFT Klik disini untuk NPLCTweb
2	Penyerapan Karbon dan Produksi Oksigen di Tegakan Atas di Arboretum PT Arutmin Indonesia NPLCT	1. *International Journal of Social Service and Research* (IJSSR) 2. Arutmin NPLCT CSR Enviro	1. Akademisi 2. LSM 3. Pemerhati Lingkungan 4. Publikasi ke Publik	Klik disini untuk IJSSR1 Klik disini untuk NPLCTweb
3	Perubahan Tingkat Kehijauan Di Kawasan RPT NPLCT PT Arumin Indonesia Dengan *Normalized Difference Vegetation Index* (NDVI)	1. *International Journal of Social Service and Research* (IJSSR) 2. Arutmin NPLCT CSR Enviro	1. Akademisi 2. LSM 3. Pemerhati Lingkungan 4. Publikasi ke Publik	Klik disini untuk IJSSR2 Klik disini untuk NPLCTweb

4	PT Arutmin Indonesia Berkontribusi di Kegiatan HBS 2024, Minimalkan Penggunaan Plastik	Arutmin NPLCT CSR Enviro	1. Publikasi ke Instansi 2. Pemerintah (Dinas Lingkungan Hidup) 3. Publikasi ke Masyarakat Lokal	Klik disini untuk lihat berita
5	*Mangrove For Banua* dari Arutmin Nplct	Arutmin NPLCT CSR Enviro	1. Publikasi ke Instansi 2. Pemerintah (Dinas Lingkungan Hidup) 3. Publikasi ke Masyarakat Lokal	Klik disini untuk lihat berita
6	Program Kali Bersih Wujud Kepedulian PT Arutmin Terhadap Lingkungan	Arutmin NPLCT CSR Enviro	1. Publikasi ke Instansi 2. Pemerintah (Dinas Lingkungan Hidup) 3. Publikasi ke Masyarakat Lokal	Klik disini untuk lihat berita
7	Tiung, Wisatawan Dimanjakan dengan Berbagai Fasilitas ini	Arutmin NPLCT CSR Enviro	1. Publikasi ke Instansi 2. Pemerintah (Dinas Lingkungan Hidup) 3. Publikasi ke Masyarakat Lokal	Klik disini untuk lihat berita

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Mapping Of Subtrate Categories With The OBIA Method Using SPOT-7 Imagery On Fringing Reef At Tanjung Pemancingan, Kotabaru Regency

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Abstract:
Background*: Tanjung Pemancingan has a fringing coral reef ecosystem close to mangrove ecosystems and river mouths. Coral reefs in this region are subject to both anthropogenic and anthropogenic factors. These factors affect the condition of the coral reef substrate.*
Materials and Methods*: This study aims to map the categories of coral reef substrates using the OBIA method and the SVM algorithm.*
Results*: The results showed that for the classification of the ten categories of substrates, the OA (overall accuracy) was 65.35%. The results of the classification categories of 10 substrate categories show that the biotic HC (hard coral) category has an area of about 9 ha, the coral category mixed with other substrates (HC Mix) is around 0.55 Ha, the SP (sponge) the substrate category is around 0.17 Ha and SC (soft coral) about 0.10 Ha. Furthermore, DCA mixed or dead coral covered by algae and mixed with other substrates with other substrate categories around 4.65 Ha, followed by DCA around 3.51 Ha and DC (dead coral) around 0.1 Ha. While abiotic elements such as sand (S) are around 5.41 Ha, then HG (hard ground), which is dead reef flat, covered by sediment, and exposed at the lowest tide, has an area of around 50 ha. Another abiotic category, RCK-S, covering an area of 7.06 Ha, is a category of andesite rock and sand outcrop substrates near coral reefs and beaches.*
Key Word*: Coral Reef Mapping, SPOT-7 Imagery, Substrate Classification, OBIA, SVM, Tanjung Pemancingan*

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I. Introduction

An exotic landscape characterizes Tanjung Pemancingan Kotabaru, and its coastal area has mangrove ecosystems and fringing-type coral reefs. These marginal coral reefs are inseparable from the supply of coral larvae originating from the Makassar Strait, especially the Sunda barrier reef region distributed by the current southward in a stepped manner stone from island to island or through connectivity between coral reefs.

The distribution of coral reefs on the west side of Tanjung Pemancingan is adjacent to mangrove vegetation. This condition is possible due to the change in the mass of water from the Makassar Strait, which brings more transparency and more salinity water during high tide. Currents and those that carry more transparent water allow the cleaning of sediment on the coral surface and carry natural food in the form of plankton.

Coral reefs in the Tanjung Pemancingan area play a significant role in maintaining coastal stability from abrasion. The fringing coral reefs around the coast play a role in dampening wave energy. This ecosystem provides a diverse source of germplasm, alternative food and pharmaceutical sources, and potential environmental services that can be utilized in the future.

On the other hand, climate change marked by La Nina and El Nino phenomena also affects coral reef ecosystems. When floods are triggered by high to extreme rainfall, causing pools of freshwater flood coastal areas. At the same time, low-salinity and turbid water also spread in coral reef waters.

El Nino events are characterized by an anomaly in seawater temperature where an increase in temperature above the normal average also causes coral bleaching, resulting in death. This event also occurred in 2010, mid-2015 to early 2016. Likewise, at the end of 2019 and early 2020 at Tanjung Pemancingan, there was partial bleaching of the fringing coral reefs and coral reefs in the breakwater. In addition to natural events, other challenges include damage due to anthropogenic factors, both directly and indirectly.

In addition to natural events, other challenges include damage due to anthropogenic factors, both directly and indirectly. Increased sedimentation in coastal areas results from forest conversion in the upper region, which

has increased runoff during the rainy season. This sedimentation is always followed by an increase in nutrients in coral reef waters, which impacts increasing benthic algae and sponges replacing coral formations. Likewise, organic and plastic waste increase negatively impacts the coral reef ecosystem.

According to Burke et al. (2012), South Kalimantan, especially the coral reefs in the Kotabaru Regency area, are classified as moderate to high based on the combined level of local threats. The combined local threat level includes coastal development, watershed-derived pollution, pollution and damage stemming from marine use patterns such as overfishing and destruction.

Post-mine closure of coal mines, including coal ports and terminals, will be implemented in the next two decades. PT Arutmin Indonesia NPLCT plans that the Tanjung Pemancingan area will be used as a micro-tropical forest area, including mangroves and coral reefs as an ecotourism and environmental education, developing a Silvio fishery and turning coral reef waters into a tourism and educational area. All of this aligns to achieve the SDGs, namely a maintained environment and poverty alleviation through community development.

In order to make a plan for managing the reef ecosystem in the Tanjung Pemancingan area, it is necessary to prepare a database which begins with mapping activities with object-based analysis of satellite imagery. The mapping and analysis aim to see in detail the substrate's distribution, characteristics, and extent of coral reefs.

II. Materials And Methods

Data collection activities were carried out during September and November 2021. This activity is located around the port operation area and coal terminal of PT Arutmin Indonesia NPLCT. The research area covers the fringing reef area along the east side to the north of the Kotabaru Tanjung Pemancingan. Starting from the flat fringing reef around Bangko Sarang Tiung bay to the north of the NPLCT loading jetty and its west side. A map of data collection locations can be seen in Figure 1.

The tools and materials used in this study include diving equipment, slates and pencils, underwater cameras, GPS (Global Positioning System, roll meters, 1.5 x 1.5 m plots, laptops, Microsoft Office 2019, ArcGIS Desktop 10.8, ENVI 5.3, ER Mapper 2015, eCognition Developer 64 bit version 9.1 and Surfer 1, SPOT-7 2020 Imagery.

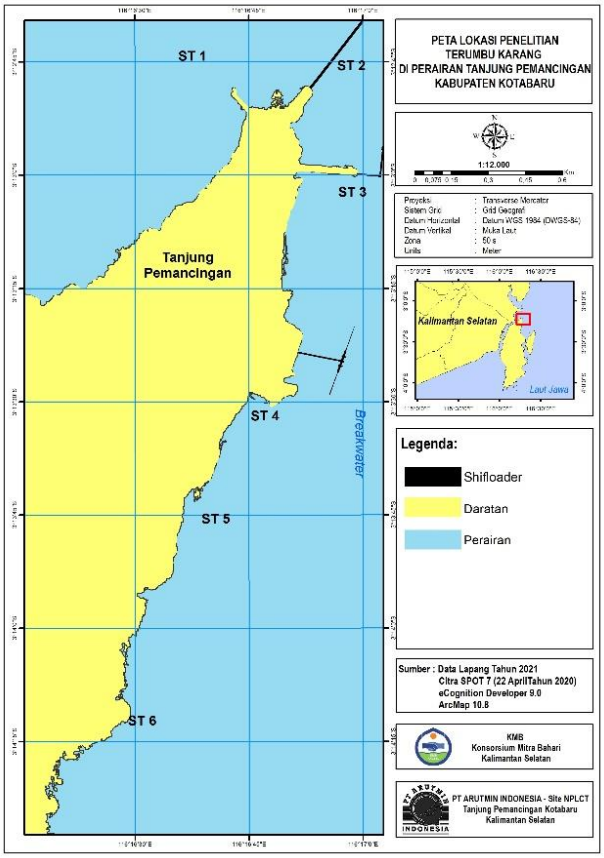

Figure 1. Data Collection Locations

Data Acquisition

Sampling Location Determination

Coral reef data collection locations were carried out using purposive sampling, taking as many as six stations for squares or plots on the reef flat to the edge on the east and north sides of Tanjung Pemancingan.

Observation and Measurement of Water Floor Objects

Field data collection is needed to complete image analysis using a systematic random sampling method. The distance between sampling points is around 5 – 20m depending on homogeneity and heterogeneity. The observation locations were determined when planning the field survey, starting with the visual interpretation of Spot 7 imagery recorded in early 2020. Data collection considered shallow water benthic cover diversity and the spatial resolution of the satellite images used in the classification. The field observation process uses a 1.5 m2 (plot) to assess substrate cover and Hand GPS to record coordinate data. The use of underwater cameras for plot documentation facilitates post-processing and categorizing data sets.

In this study, the quadratic (plot) method was used to verify the results of image analysis with a focus on assessing the percentage cover of the substrate category in the plot of the observation area. Observation of this bottom substrate using snorkelling and diving techniques in one plot 1.5 x 1.5m. The area of the plot is adjusted to the spatial resolution of the satellite image used in the classification to assess the benthic category.

The assessment of the substrate category on coral reefs refers to the Global Coral Reef Monitoring Network (GCRMN) assessment guide and English et al. (1994), which focused on ten categories of 31 categories of substrates, namely Biotic (hard coral, soft coral, sponge, other fauna, benthic algae), Abiotic (sand and rubble) and dead Scleractinia (dead coral and dead coral with algae).

Data Analysis

Initial Processing of Satellite Imagery

The initial processing stage of satellite imagery (image preprocessing) is carried out to improve the original image data (raw data) into satellite images ready for interpretation. The work carried out includes correcting errors due to the scattering of particles in the atmosphere recorded by satellite imagery (radiometric correction), repairing errors in the position of satellite image recording against earth references (geometric correction), masking and cropping, image composites and object sharpening in images through spectral value stretching. Image. Furthermore, water column correction is carried out to improve image quality by reducing interference in the water column.

There are two ways of correcting the water column, especially in finding the value (ki/kj), representing the attenuation coefficient ratio. First, through the gradient values on the linear lines formed by a pair of visible spectrum bands. Second, it can also be obtained based on the Lyzenga formula (1981) as follows:

$$Y = (\text{In } band\ 1) + \left(\frac{ki}{kj} \times \text{In } band\ 2 \right)$$

$$\frac{ki}{kj} = \alpha + \sqrt{a^2 + 1}$$

$$\alpha = \frac{(Varian\ band\ 1 - Varian\ band\ 2)}{(2 \times covariant\ band\ 1\ dan\ band\ 2)}$$

Ki/kj = Ratio of the blue and green channel coefficients.

Object-Based Image Analysis (OBIA)

The object-based classification was done to recognize objects based on groups of pixels, not just on individual pixels. This technique is known as Object-Based Image Analysis (OBIA). This classification uses an algorithm to classify pixel values for proximity to pixels in a specific class based on a sample of pixels. The image resulting from the guided classification algorithm is considered an intricate classification image. After that intricate image, The results are smoothed based on the polygon boundaries or areas formed by the previous image segmentation process.

Support Vector Machine (SVM)

Additional field information in the form of benthic habitat cover for classification using the Support Vector Machine (SVM) algorithm. The SVM concept tries to find the best separating function among the unlimited number of functions. Six best separator functions can be found by measuring the separator limits and looking for the maximum value. Wahidin et al. (2015) in Manan (2019) stated that the SVM algorithm could produce better accuracy in OBIA-based methods. Mathematically, the following equation can use the basic concept of SVM based on (Vapnik, 1982 in Manan, 2019)

$$K = \exp\left(-\frac{\|x-x'\|^2}{2\sigma^2}\right) \quad \ldots(1)$$

Where: X represents the vector of each data, represents the number of degrees of the polynomial function.

Validation/Accuracy Test

The accuracy test carried out in this study was to compare the classification results of benthic distribution maps with field data. The method used in this accuracy test is a confusion matrix based on the following equation (Congalton and Green, 2008 in Manan, 2019):

Overall Accuracy (OA) : ..... (2)

$$OA = \frac{\sum_{i}^{k}=1\, n_{ii}}{N}$$

Producer Accuracy (PA) : ..... (3)

$$PA = \frac{n_{ii}}{n_{i+}}$$

User Accuracy (UA) : ..... (4)

$$UA = \frac{n_{ii}}{n_{+i}}$$

Description:

N : the number of objects contained in the classification and reference results

n_{i+} : the total number of objects in the i-th row

n_{+i} : the total number of objects in the ith column

n_{ii} : the diagonal value of the matrix of the i-th row and i-column

The acceptable accuracy limit for shallow water bottom habitat maps based on SNI 7716:2011 concerning Mapping shallow sea bottom habitat is 60%.

III. Result

Water Column Correction

The water column correction is used by calculating DII (Deep Invariant Index) from the Lyzenga formula which uses information from the Ratio of the attenuation coefficient for each pair of visible light bands. The results of the calculation of the attenuation coefficient ratio of the visible light band pair of SPOT 7 images are presented in the following table:

Table 1. The Ratio of the attenuation coefficient of the visible light band pairs of the SPOT 7 image

Band Combination	Attenuation Coefficient (ki/KJ)
DII 2/3 (blue - green)	0.010752253
DII 2/4 (blue-red)	0.001771271
DII 3/4 (green - red)	0.003958538

Creating a Classification Scheme

The naming and description of the ten substrate category components are based on the dominant coverage of the substrate category at each observation point, composed of one or more substrate category components. The naming, description and code for each class of substrate category are presented in Table 2.

Table 2. Name and Code of Substrate Category

Substrate category class	Class Code	Description
Hard Corals	HC	*Hard corals* predominate
Hard Coral Mix	HC Mix	Mixed *hard corals*
Dead Coral with Algae	DCA	Dominant *Dead Coral with Algae*
Dead Coral with Algae Mix	DCA Mix	*Dead Coral with Algae* mixed
Dead Coral	DC	Dominant *Dead Coral*
Sand	S	*Sand* dominant
Sponges	SP	*Sponge* dominant
Hard Ground	HG	*Hard Ground* dominant
Soft Corals	SC	Dominant *Soft Coral*
Rock – Sand	RCK S	*Rock* mix *Sand*

The naming of the class of substrate categories in this study was adjusted to the composition of the dominant substrate category observed in the field because the classification scheme for the classification of substrate categories does not have standard provisions or standardization in terms of mapping.

The classification scheme produced in this study consists of 2 levels of classification schemes (Figure 2). The reef level classification scheme (level 1) consists of 3 classes, namely the land/cloud class, shallow marine waters, and deep sea waters. In comparison, the substrate category classification scheme (level 2) consists of 10 substrate category classes (Table 3). The classification scheme developed in this study may be different from the classification scheme for shallow water substrate categories in other places, considering that the complexity of the substrate categories varies from region to region and this is strongly influenced by many factors.

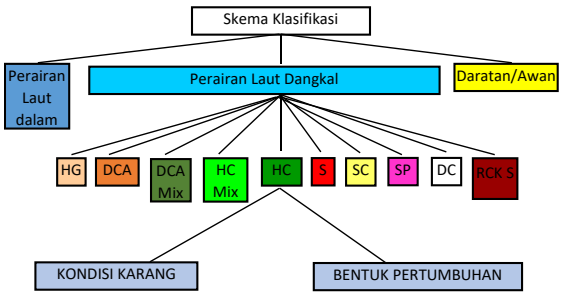

Figure 3. The results of the 2-level classification scheme

The classification scheme for shallow sea water substrate categories resulting from 250 observation points will then be divided into two, namely, 185 observation points will be used as RoI (region of interest) data in the image classification process. Then the remaining 65 observation points will be used as data for testing the accuracy of the image classification results. Classification schemes for the ten resulting shallow water substrate categories will all be used in the SPOT 7 image classification process.

Image Classification

The image classification used in this study is object-based classification (OBIA), and the image used is SPOT 7 imagery. Segmentation is the basis of the object-based classification process to produce segments or objects which will then be assigned to a specific class in the classification process.

Segmentation Scale Optimization

The segmentation process in this study uses the multiresolution segmentation (MRS) algorithm. Segmentation by applying several different scales (multiscale) can produce objects with different numbers and shapes based on the complexity of an object. At level 1, the segmentation scale used is 100 to produce 9,765 objects. Based on the segmentation process of scales 2, 3, 5, and 10 in the SPOT 7 imagery, the number of objects was 691,746, 350,701, 133,538, and 33,711, respectively. Wahiddin (2015) has proven that the effect of segmentation scale can affect the shape, size, and number of objects produced.

Reef Level Classification (Level 1)

Reef level classification is the basis for the classification process for shallow water substrate categories (level 1). Image classification at level 1 uses the contextual editing method. Classification level 1 in this study resulted in 3 classes: cloud/land (land), shallow sea waters (Area of Interest) and deep sea waters.

Classification results of level 1 are processed from objects that are formed on the results of segmentation, namely, forming 9,765 objects. From the total area of the research location of 9,776 ha, the area of each class was obtained based on the classification results, namely on the image obtained land class of 1,062.66 Ha, shallow sea water class of 323.93 Ha and deepsea class of 1,133.22 Ha, obtained area of each class and maps based on the classification results as follows:

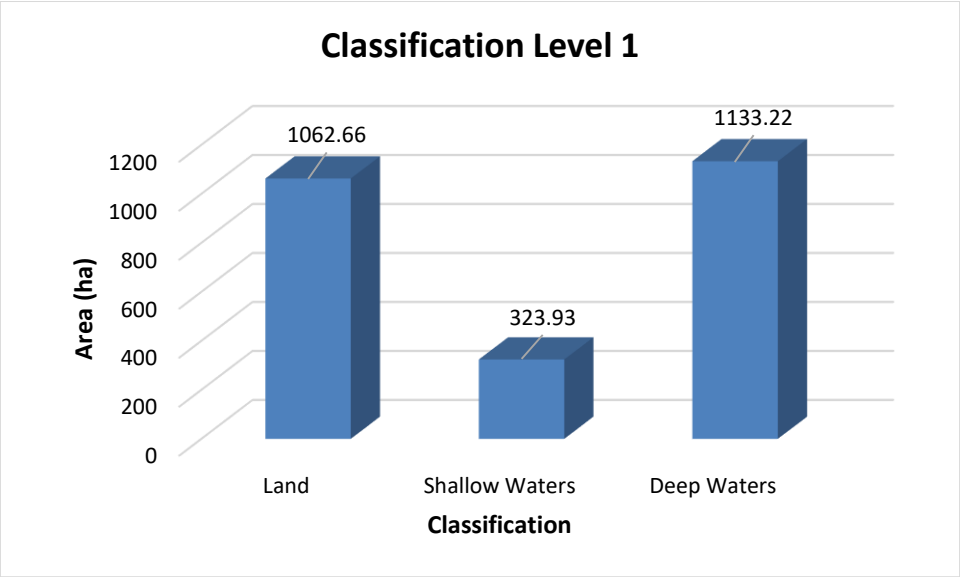

Figure 4. Class area of level 1 Classification Results

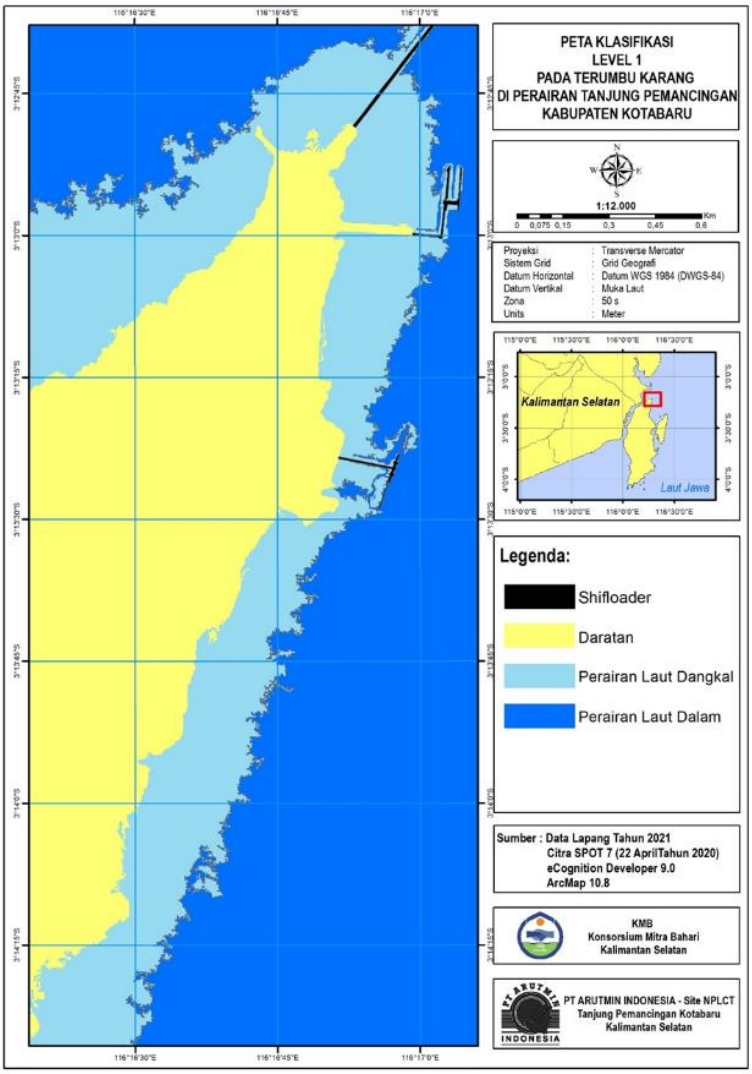

Figure 5. Class Area of level 1 Classification Results

Substrate Category Classification (Level 2)

The results of level 1 classification, especially in the shallow sea waters class, are used as area boundaries in classifying substrate categories for level 2 classification. The shallow sea waters class at level 1 is then re-segmented at level 2 by applying optimization of segmentation scales 2, 3, 5, and 10.

The resulting objects or segments are then classified by guided classification using a machine learning algorithm, namely SVM, using data from the classification scheme for shallow water substrate categories (Table 3) as input thematic layers that have been made previously based on direct observations in the field. From a total of 250 field observation points, 185 observation points are used as input data (input thematic layer) in the classification process, and the remaining 65 observation points are used as data to test accuracy with the confusion matrix. The feature input in the level 2 classification process with SPOT 7 images uses the feature input, namely the layer values (mean and standard deviation) of all visible light bands and the results of the three DII band pair compositions.

The accuracy test results using the SVM algorithm on the segmentation scale 3 obtained PA (Producer Accuracy) and UA (User Accuracy) values ranging from 28.57% - 100%. The complexity of the substrate categories strongly influences the low accuracy value in several classes of substrate categories at the research location and the number of observation points in each class of substrate categories. In addition, the classification algorithm cannot avoid the spectral similarity between the substrate classes, especially for classes composed of two components of the substrate category. According to Anggoro et al. (2017), the factors that affect low accuracy are due to the high complexity of the substrate categories in the study area and the equivalence between GPS accuracy and the spatial resolution of the images used. Green et al. (2000) stated that the accuracy of the substrate category mapping can be used if the substrate category classification map produces an overall accuracy (OA) of > 60%, meaning that the accuracy of the substrate category map produces an OA (Overall Accuracy) value below the number it cannot be used for analysis or other purposes and in this study the accuracy test for primary substrates refers to SNI 7716: 2011, which is an accuracy rate of 60% (BIG, 2014). Based on this, in this study, the results of the classification of shallow water substrate categories can be used considering the accuracy test results yield an overall accuracy value above 60%, namely an OA of 65.35% using the SVM algorithm on a segmentation scale of 3.

In the calculation of the accuracy test for the classification results, the amount of field data used to test accuracy with the amount of data displayed in the confusion matrix table is different from the amount of data because the accuracy test calculation table is the result of recognition software calculations where the amount of data increases due to the process of segmentation/grouping data. The following, are the results of the accuracy test of the level 2 classification:

Table 3. Producer Accuracy, User Accuracy and Overall Accuracy for Ten Benthic Categories

Field / Image	HG	HC	DCA mix	SP	SC	DCA	S	HC Mix	DC	Sum	User Accuracy
HG	**6**	0	0	0	0	0	4	0	0	10	60
HC	0	**31**	2	4	3	3	2	1	0	46	67,39
DCA mix	1	0	**4**	0	0	1	0	0	0	6	66,67
SP	0	0	0	**2**	0	0	0	0	0	2	100
SC	0	0	0	0	**10**	0	0	0	0	10	100
DCA	0	2	0	0	0	**5**	3	0	0	10	50
S	1	1	1	0	0	1	**4**	2	1	11	36,36
HC mix	0	1	0	0	0	0	1	**3**	0	5	60
DC	0	0	0	0	0	0	0	0	**1**	1	100
Sum	8	35	7	6	13	10	14	6	2	**101**	
Producer Accuracy	75	88.57	57,14	33,33	76,92	50	28.57	50	50	OA	**65,35**

The results of the classification of substrate categories (level 2) using the SVM algorithm are presented in Figure 6 for the application of the ten classes of substrate categories and Figure 7 for the percentage diagram of the area covered by the substrate categories in the ten classes of substrate categories, presented as follows:

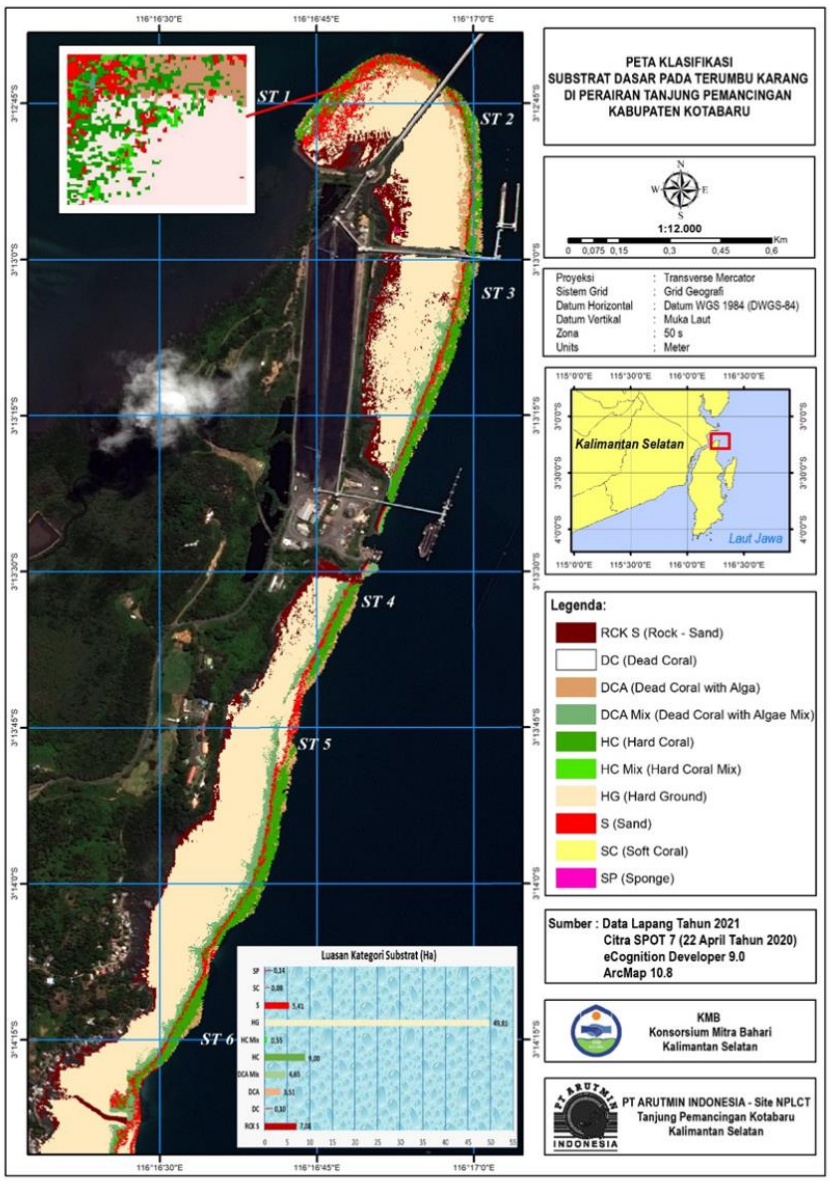

Figure 5. Classification Map of 10 Classes of Substrate Categories on Fringing Reefs at Tanjung Pemancingan Using a Scale of 3 on SPOT 7 Imagery

Based on the results of the bottom water substrate classification map (Figure 6), it is clear that the substrate categories are distributed in shallow water areas. Visually, it can be seen that there are differences in the distribution of benthic habitat classes (10 classes). This difference is due to the difference in the number of classes formed, which will affect the classification process and will undoubtedly have an impact on the results of the accuracy of the water bottom substrate classification process.

From the results of this classification, the area of each of the ten classes of sub-watery substrates at the research location can be obtained using the SVM algorithm with a segmentation scale of 3 in the SPOT 7 image.

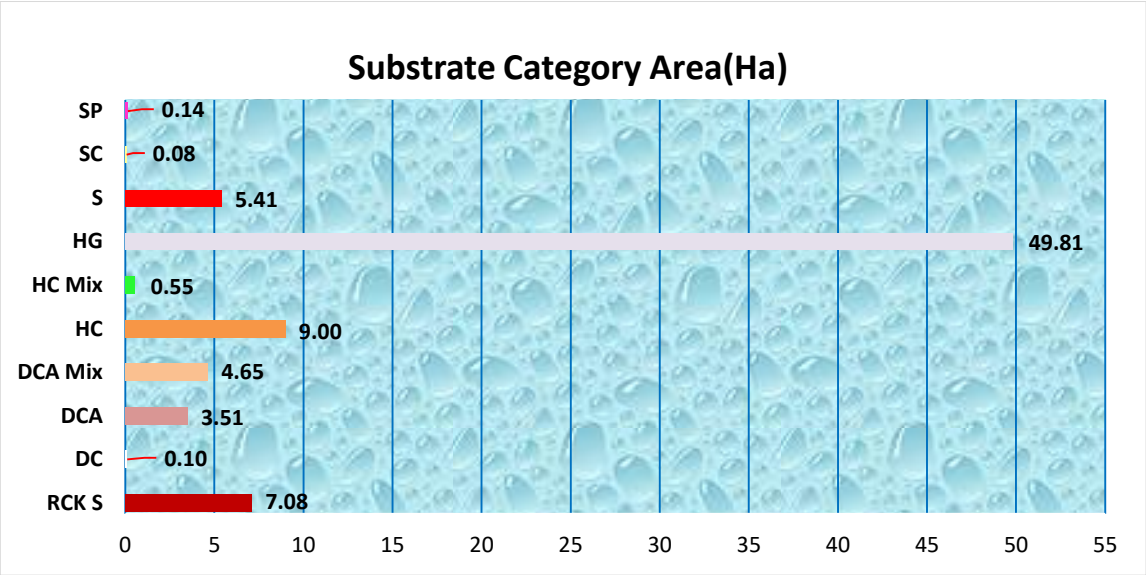

Figure 6. Distribution of the Area of 10 Classes of Substrate Categories as A Result of Classification with The SVM Algorithm on The Segmentation Scale 3 SPOT Image 7

Based on Figure 6, it can be seen that the results of the classification of substrate categories in 10 classes show that the HG (hard ground) class dominates the shallow water area at the study site, namely an area of 50 Ha. The dominant HG class shows the area behind the reef crest, which appears at low tide, directly exposed to air, heat, rainwater and accumulated runoff flows. Class HG is a hard flat limestone or rock material covered by a thin layer of sand and silt sediments. In contrast, the class of substrate category with the smallest area is obtained in the SC and DC classes of 0.08 Ha and 0.10 Ha, respectively.

IV. Conclusion

Analysis of SPOT 7 satellite imagery using the OBIA (object base image analysis) classification method with the SVM (support vector machine) algorithm shows that the classification of 10 categories of substrate classes having an OA (overall accuracy) of 65.35% already fulfils the minimum SNI accuracy test criteria of 60%. The calcification map of 10 substrate categories shows that the biotic category HC (hard coral), which is coral has an area of about 9 Ha, the category of coral mixed with other substrates (HC Mix) is around 0.55 Ha, the substrate category SP (sponge) is around 0.17 Ha and SC (soft coral) around 0.10 Ha. Furthermore, DCA mix or dead coral covered by algae and mixed with other substrates with other substrate categories around 4.65 Ha, followed by DCA around 3.51 Ha and DC around 0.1 Ha. While abiotic elements such as sand (S) are around 5.41 Ha, then hard ground (HG), the flat reef that is dead and covered with sediment, has an area of around 50 Ha. Another abiotic category, RCK-S covering an area of 7.06 Ha, is a category of andesite rock and sand substrates near coral reefs and beaches.

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Sequestration of Carbon and Oxygen Production in Upper Stands in The Arboretum of PT Arutmin Indonesia NPLCT

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Keywords
Biomass, Carbon stocks, Carbon sequestration, Oxygen, Arboretum.

ABSTRACT
This study aims to determine biomass, carbon, carbon uptake, and oxygen production in the arboretum of PT. Arutmin Indonesia-NPLCT. The methods used are purposive sampling and allometrics. The plot size useda is 20 m × 20 m with 12 plots measured. The area of the arboretum is 5.05 ha. Carbon stock measurement is done by multiplying biomass by a conversion rate of 0.47 (47%) while carbon dioxide (CO2) sequestration is calculated by multiplying the average annual biomass growth by the conversion rate of 1.4667 obtained from the photosynthesis equation and oxygen production is measured based on conversion from carbon yield. Based on the results of the study, it showed that the biomass content of stands in the arboretum was 1,067.32 tons, carbon stocks were 501.64 tons, carbon dioxide (CO2) absorption was 1,565.45 tons, and oxygen production was 1,337.71 tons.

INTRODUCTION

Indonesia's forests are the richest and most diverse forests on earth. The Indonesian government owns more than 120 million hectares of secret forest in the country, almost twice the size of France. Of the 4,444,120 million hectares, Indonesia's forests are divided into three categories. This means that 18% is parkland, 25% is conservation forest, and 57% is production forest. A total of 69 million hectares of forest is designated for construction. However, only 45% of land is granted permits to private companies as natural production forests or plantation forests. The remaining 55% of productive forests are outside the lease system and are particularly vulnerable because there are no institutions to guarantee the management of these forests.

Trees or forests are the world's lungs since numerous trees have become one of the biggest oxygen makers (Bandara & Dissanayake, 2021; Han et al., 2023; Yang et al., 2023). Where oxygen could be an essential requirement in human life. Woody vegetation, especially trees, provides many benefits that can improve the quality of the environment and human health in and around urban areas (Batkhuyag et al., 2023; Chaudhary & Aryal, 2023; Kala, 2023; Miranda Verly et al., 2023). These benefits include improved air and water quality, energy savings in buildings, cooler air temperatures, reduced UV radiation, and many other environmental and social benefits (Nowak et al., 2007). Fast-growing woody plants can absorb more carbon than slow-growing plants. Faster-growing vegetation often has difficulty measuring its ability to sequester carbon (Kongboon et al., 2022; Singh et al., 2023; Swamy et al., 2023; Wang et al., 2024). Forest plays a role in increasing CO_2 absorption through sunlight and soil water. Chlorophyllous plants can absorb CO2 from the atmosphere (Kumari et al., 2023; Mohammadi et al., 2023; Sribianti et al., 2022).

Through photosynthesis, the results are among other things, stored as biomass, allowing plants to grow larger or taller. This growth will continue until the vegetation stops growing physiologically or is harvested (Song et al., 2024; Wu et al., 2023; Zhou et al., 2023). With sustainable

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forests, more carbon (C) is stored and longer. Therefore, planting vegetation on bare land or restoring destroyed forests will help absorb excess CO2 in the atmosphere (Bajuin, 2018; Birungi et al., 2023; Namin et al., 2020).

The distinctive focus of this research centers on the Arboretum of PT. Arutmin Indonesia-NPLCT is an ecological setting characterized by its unique vegetation composition and environmental conditions. While existing literature acknowledges the general importance of forests in carbon absorption and oxygen production, this study seeks to contribute to the existing body of knowledge by unraveling the intricacies of these processes within the specific context of an arboretum dedicated to reforestation and ecological restoration.

Unlike broader studies that encompass carbon absorption in various forest ecosystems, this research scrutinizes the effectiveness of reforestation efforts. Doing so provides insights that may diverge from findings in other forested areas. Moreover, the study may integrate innovative methodologies and technologies, such as advanced sensor technologies or spatial mapping, thereby enhancing measurement precision and setting itself apart from previous research endeavors.

By explicitly highlighting these distinctive aspects, the research positions itself as a valuable addition to the scientific discourse on forest ecosystems and emphasizes its potential to refine environmental management strategies. This is particularly relevant in sustainable practices within arboretums, underscoring the significance of the study's potential contributions to advancing ecological restoration and management practices.

METHODS

The research was conducted at the Arboretum of PT. Arutmin Indonesia-NPLCT,Jl. Berangas Km. 5,5 Tanjung Pemancingan Sarang Tiung village, Kotabaru Regency, South Kalimantan. The time required to carry out this research is 2 months.

The tools used are GPS (Global Positioning System), cameras for documentation tools, meters to measure distances, measuring tapes to measure diameters. The object in this study, is the stand on the tree contained in the Arboretum of PT. Arutmin-NPLCT.Tree Measurement at the study site. The data collected or obtained from the field is diameter. The biomass formula used is:

Top Biomass Measurement

Table 1. Biomass Calculation Formula

Types of Vegetation	Formula	source
Acacia	$Y=0{,}060255\times D^{2{,}39}$	Purwitasari (2011)
Tropics Common (Branching Tree)	$Y=0{,}11\times\rho\times D^{2{,}62}$	Ketterings et,al., (2001)
Palm <1,3 m	Y = 0,0976 H+0,0706	ICRAF (2009)
Palm >1,3 m	Y = 0,002382 × D 2,3385 × H0,9411	Lubis (2011)

Information:
Y = Biomass (kg)
H = Tree Height (m)
D = Diameter at Chest Level (dbh) (cm)
ρ = Wood Density/Specific Gravity (mg/m^3, kg/m^3, atau g/m^3)

Bottom Biomass Measurement

Measurement of biomass under the root part of the plant. The formula used is:

$$Bb = Ba \times 0{,}25$$

Information:
Bb = Bottom Biomass (kg)
Ba = Top Biomass (kg)
0,25 = Constant Value (IPCC,2006)

Carbon (CO_2) From Biomass
Calculation by Formula (IPCC, 2006) is:

$$C = \text{Biomass} \times \%\text{C-Organik}$$

Information:
C = Carbon from Biomass (kg)
B = bioresource Total (kg)
%C Organik = 0,47% (SNI 7724, 2011)

Reserve Carbon for Ha for Plot
The carbon value per ha for data on the soil surface is:

$$Cn = \frac{\text{Cx}}{1000} \times \frac{10.000}{Lplot}$$

Information:
Cn = Carbon pool on plot (ton/ha).
Cx = Carbon pool on plot (kg).
Lplot = Area of plots in each carbon pool (m2) (SNI 7724, 2011)

Sequestration Karbon (CO_2)
Sequestration CO_2, rumus ialah:

$$\text{Sequestration } CO_2 = \text{Biomass (kg)} \times 1{,}4667$$

Information
1,4467 = The conversion value obtained from the photosynthetic equation (Baharuddin et al., 2014).

Conversion of Carbon Biomass to Oxygen (O_2)

1. **CO_2- equivalent**
   The calculation uses a ratio of the relative atomic mass of C (12) and the passive molecular mass of CO_2 (44) with the formula (Kemenhut,2013) :

$$CO_2 - ekuivalen = (\frac{44}{12}) \times \text{C}$$

2. **Oxygen Production (O_2)**
   Oxygen production/release is calculated based on the amount of oxygen produced during photosynthesis minus the amount used during plant respiration. The amount of oxygen production can be estimated from carbon sequestration based on atomic weight (Ross and Salisbury, 1978):

$$Net\ O^2 = net\ C \times \frac{32}{12}$$

The determination of carbon storage measurement plots is carried out with the following sizes: Plot size: "20m × 20m."

Sampling using purposive sampling and sampling methods without harvesting or logging. This sampling method involves measuring the height of oil palms and tree diameters by applying allometric equations to extrapolate biomass. Determination of the number of tiles that the measurement takes, using 10% IS. Based on Boon and Tideman (1950) for forests whose area ≥ 1,000 ha IS used 2%, while the area < 1,000 ha IS used 5% to 10%. Based on the area in the Arboretum of PT. Arutmin NPLCt is 5.05 ha.

The parameters measured are (dbh ≥ 10 cm and ≥ height 1.5 m) by measuring the diameter of the tree stand. Form a stand plot like Figure 1.

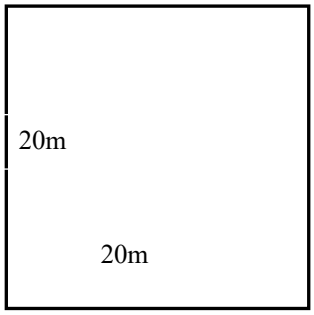

Figure 1. Plot Size Shape

RESULTS

Based on stand diameter measurements that have been carried out in the arboretum of PT. Arutmin Indonesia-NPLCT. The results obtained are biomass average 211.351 tons/ha, average carbon 99.33 tons/ha, carbon absorption 309.98, average CO2-equivalent 364.22, and average oxygen production 264.894 tons/ha. The results of these results can be seen in Table 2.

Table 2. Total Biomass, Carbon Stocks, Carbon Sequestration, CO2-equivalent, and Oxygen Production in Arboretums

No.Plot	Biomass Total (Ton/Ha)	Carbon Total (Ton/Ha)	Sequestration Carbon (Ton/Ha)	CO2-Ekuivalen (Ton/Ha)	Production Oksigen (Ton/Ha)
1	270,75	127,25	397,10	466,59	339,33
2	11,58	5,44	16,98	19,95	14,51
3	93,50	43,95	137,14	161,14	117,19
4	153,72	72,25	225,46	264,91	192,66
5	277,86	130,59	407,53	478,84	348,25
6	28,42	13,36	41,68	48,97	35,62
7	307,72	144,63	451,33	530,30	385,67
8	141,64	66,57	207,74	244,09	177,52
9	314,84	147,98	461,78	542,58	394,60
10	594,16	279,26	871,46	1023,94	744,68
11	153,88	72,32	225,69	265,18	192,86
12	188,16	88,43	275,97	324,26	235,83
Average	**211,351**	**99,335**	**309,989**	**364,229**	**264,894**
TOTAL	**2536,22**	**1192,02**	**3719,87**	**4370,75**	**3178,73**

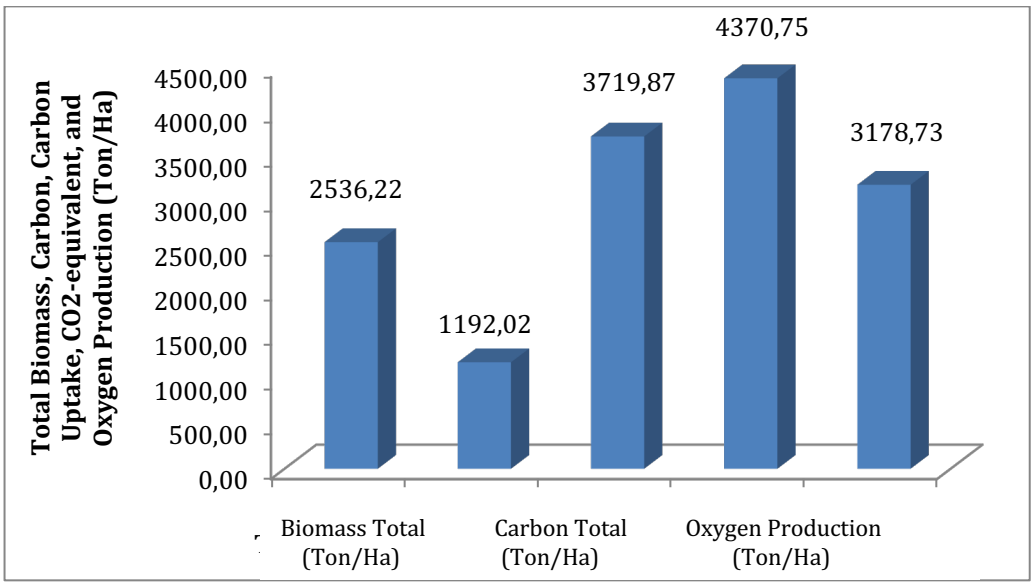

Figure 2. Diagram of total biomass, carbon, carbon sequestration, CO2-equivalent, and oxygen production (ton/ha)

Arboretum located in the area of PT. Arutmin Indonesia-NPLCT covering an area of 5.05 ha. The results obtained for total biomass, total carbon, total uptake, CO2-equivalent and total oxygen production in the area can be seen in table 3.

Table 3. Total Biomass, Carbon Stocks, Carbon Sequestration, CO2-Equivalent, and Oxygen Production in an Arboretum covering an area of 5.05 ha

No	Arboretum Area (Ha)	Average Biomass (Ton/Ha)	Average Carbon (ton/ha)	Average Carbon Sequestration (ton/ha)	Average CO2-Ekuivalen (ton/ha)	Average Oxygen Production (ton/ha)
1	5.05	211,35	99,34	309,99	364,23	264,89
		Biomass Total (Ton)	Carbon Total (ton)	Carbon Sequestration (Ton)	CO2- Ekuivalen (Ton)	Oxygen Production (Ton)
2		1.067,32	501,64	1.565,45	1.839,36	1.337,71

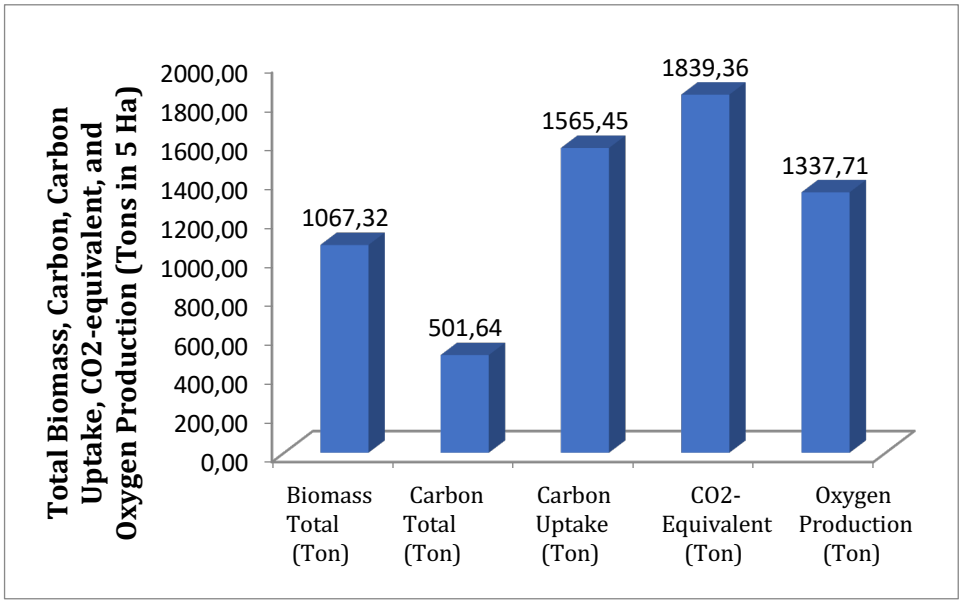

Figure 3. Diagram of total biomass, carbon, carbon sequestration, CO2-equivalent, and oxygen production in an area of 5.05 ha (tons)

Based on figure 3, it can be seen that the results obtained from the calculation of the biomass of tree stands in the arboretum covering an area of 5.05 ha. Carbon produced amounted to 501.64 tons, carbon absorption amounted to 1565.45 tons, and oxygen production 1,337.71 tons. Based on the calculation results of oxygen production produced in the arboretum area of PT. Arutmin Indonesia-NPLCT with an area of 5.05 Ha. The oxygen production produced amounted to 1,337.71 tons. According to the global oxygen budget and its future article published in the science bulletin. Each adult needs 1.17 kg of oxygen per day. It can be known that the production of oxygen produced in the arboretum is 371,586 kg / day. So that it can meet the daily oxygen needs of adults as many as 317,000 people per day.

According to technical guidelines for planting tree species absorbing air pollutants published by the Ministry of Environment and Forestry (2015), it is said that 1 ha of green open space (RTH) filled with large trees, oxygen production produced is 0.6 tons for 1,500 residents / day and carbon dioxide absorption is 2.5 tons / year.

Based on calculations that have been done by weighing the results obtained in the average standard carbon production that has been obtained, the results exceed what has been planned using purposive and allometric methods so that what stands are measured only the stem with this purpose can be known that what is done is comparable to the greater than the average oxygen produced.

The value obtained based on what will be done with what will happen with this can be known that the gains that have been taken with what is known value by reviewing the location and measuring diamete trunks or tree stands can be seen that the results obtained are quite large for various sizes with the conditions of stands around us so that what is obtained is the same as what is in previous studies when compared The result is that the carbon production produced is very large in the arboretum area, which is 5.05 ha.

CONCLUSION

The arboretum at PT. Arutmin Indonesia-NPLCT boasts impressive ecological metrics, with an average biomass content of 211.35 tons/ha, a carbon stock of 99.34 tons/ha, carbon dioxide absorption reaching 364.23 tons/ha, and a substantial oxygen production of 264.89 tons/ha. These figures, coupled with a total biomass area of 5.05 ha, totaling 1,067.32 tons, and corresponding carbon stocks, oxygen production, and CO2 absorption, underscore the arboretum's vital role in supporting a thriving ecosystem. To better comprehend these results, it is essential to interpret them in the context of environmental significance. Comparisons with established benchmarks could provide readers with a frame of reference to evaluate the arboretum's effectiveness in biomass content, carbon stock, and oxygen production. Acknowledging study limitations, such as seasonal variations or site-specific conditions, is crucial for transparency and a more comprehensive understanding. Recommendations for optimizing carbon sequestration or biomass content could enhance the arboretum's ecological impact, making it a valuable area for future research and management strategies. In conclusion, the significant oxygen production of 371,586 kg/day positions the arboretum as a potential contributor to meeting the daily oxygen needs of around 317,000 people, emphasizing its importance in the broader environmental landscape.

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Changes In Greenness Level In PT Arumin Indonesia's RPT NPLCT Area With Normalized Difference Vegetation Index (NDVI)

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Keywords
Greenness, NDVI, Sentinel-2A, Remote Sensing, Geospatial Information System, Kotabaru Regency, South Kalimantan

ABSTRACT
The observation focused on PT Arutmin Indonesia's NPLCT RPT area of 98 hectares. These observations are viewed with the help of remote sensing technology and geographic information systems (GIS). This observation aims to analyze changes in greenness levels that occur from 2019 to 2023 using the normalized difference vegetation index (NDVI) method and the materials used sentinel-2A imagery. The goal is to evaluate changes in greenness levels and their interpolation from 2019 to 2023. An area of 98 hectares was monitored using sentinel-2A satellite imagery with a pixel sharpness of 10 meters x 10 meters. The results show sentinel imagery from 2019 to 2020, 38.41 ha experienced reforestation, 1.65 ha experienced deforestation and 58.14 ha did not change. Sentinel-2A image from 2020 to 2021, 79.09 ha experienced reforestation, 1.35 ha experienced deforestation and 17.68 ha did not change. Sentinel-2A image from 2021 to 2022, 1.08 ha experienced reforestation, 79.48 ha experienced deforestation and 17.68 ha did not change. Sentinel-2A imagery from 2022 to 2023, 0.47 ha experienced reforestation, 40.65 ha experienced deforestation and 57.17 ha did not change. The sentinel-2A image changes from 2019 to 2023, 2.27 ha reforested, 20.35 ha deforested and 75.64. These observations highlight the importance of remote sensing technology and geographic information systems (GIS) in monitoring and analyzing land cover change to support conservation, rehabilitation and natural resource management policies in Kotabaru District, South Kalimantan.

INTRODUCTION

An ecosystem unit in the form of a stretch of land containing biological natural resources dominated by trees with an environment that cannot be separated from others is

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called a forest. Certain areas designated and designated by the government to maintain their existence as permanent forests are called forest areas. This is regulated in Law Number 41 of 1999. Forest areas are divided into 3 functions, namely, production forest (HP), protection forest (HL) and conservation forest (Adinda, Rusdi, & Sugianto, 2020).

Vegetation density has decreased due to changes in land use (landuse) which can affect changes in the average surface temperature of an area. This change in land use is caused by increasing a land use from one side of the use to another which is accompanied by a decrease in other types of land use over time or it can be said to change the function of a land resulting from human activities (Wahyunto & Heryanto, 2006). The use of remote sensing data can be used to determine and map the distribution of land surface temperature. One of the data that can be utilized is Landsat satellite data using thermal sensors carried by the vehicle. Landsat is a continuation of the Landsat mission which for the first time became an earth observation satellite since 1972 (Landsat 1).

The Normalized Difference Vegetation Index (NDVI) has good accuracy in forest greenness studies. The concept of using NDVI is fairly simple, namely by utilizing the best Near Infrared (NIR) light reflected by chlorophyll and red light radiation absorbed by green plants (Akbar et al., 2020). The method of measuring canopy density is basically utilizing canopy cover. Canopy cover is the area of land covered by vertical projections of vegetation canopy or tree canopy (Jennings, Brown, & Sheil, 1999) (Wachid & Tyas, 2022).

Remote Sensing has enormous benefits of terrestrial survey activities, this is because Remote Sensing can be known the characteristics of an area without going directly into the field. This is because Remote Sensing can describe objects on the surface of the earth in accordance with the shape and location of objects that are similar to the original (Nurmalasari & Santosa, 2018). Moreover, in hard-to-reach areas, remote sensing is very effective to use. Various sectors of use such as the meteorological and climatological sector, forestry, population sector, marine (oceanography), and so on.

Deforestation is the condition of forest area that has decreased due to land conventions for infrastructure, settlements, agriculture, mining, and plantations. Forest destruction in Indonesia continues to increase every year. Forest destruction can affect climate change, because reduced forest land is one of the causes of increasing greenhouse gases, especially carbon dioxide as a source of emissions. Around 80% of emission sources are caused by forest burning (Wahyuni & Suranto, 2021).

Reforestation is the process of replanting trees on land that has previously been deforested or degraded. Reforestation helps the environment by guaranteeing or accelerating the re-establishment of healthy forest structures by regrowing the forest canopy and preserving biodiversity in ecosystems. Reforestation also helps reduce carbon emissions because trees can absorb carbon dioxide from the atmosphere through the process of photosynthesis. Land undergoing conversion needs to be reduced in order to maintain land cover (Mustafa & Marsoyo, 2020). Forest areas must be maintained so that they do not continue to decrease and be damaged. Some solutions to overcome the root of the problem of deforestation and reduce its negative impacts require reforestation activities and human awareness not to cut trees in the forest continuously (Asrin, Putri, & Utami, 2022).

The purpose of this report analyzing the level of greenness that occurs in PT Arutmin Indonesia's RPT NPLT area from 2019 to 2023. Seeing changes in the area of greenness of land that occur in PT Arutmin Indonesia's RPT NPLT area from 2019 to 2023. The benefit of this report is that it can analyze changes in the greenness level of the area or land by utilizing remote sensing science with software and satellite imagery.

METHODS

This analysis was conducted at NPLCT PT Arutmin Indonesia, Pulau Laut Utara District, Kotabaru Regency, South Kalimantan. The analysis is carried out during 3 months from preparation, collection and processing.

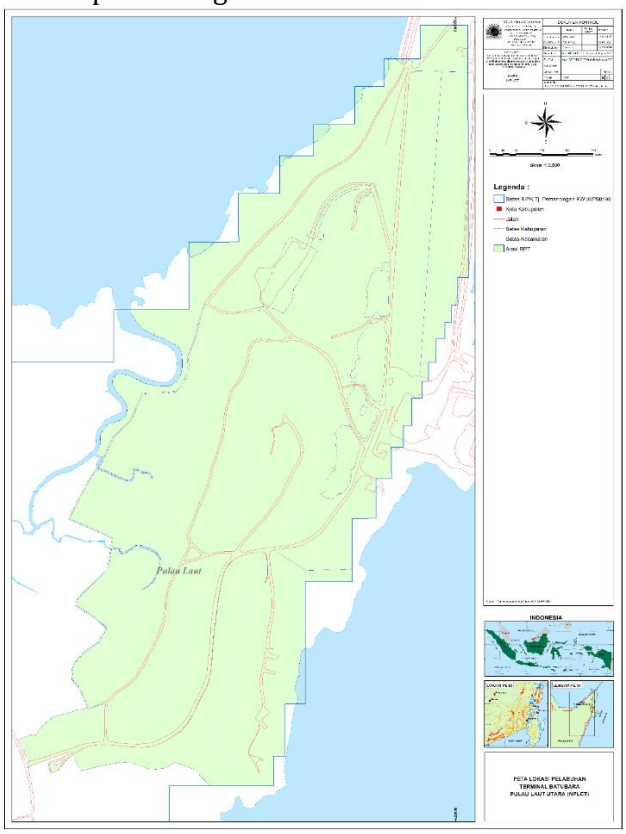

Figure 1. Location of *NDVI Analysis* at NPLCT PT Arutmin Indonesia

The object of this research is changes in the level of greenness with the *NDVI method* at NPLCT PT Arutmin Indonesia.

Tools used in the study.

1. *ArcMap Software 10.8*
2. Sentinel-2A imagery.
3. Map of Rupa Bumi Indonesia (RBI) administrative boundaries of PT Arutmin Indonesia's NPLCT working area.
4. Laptop for data processing and report writing.

Procedure

The procedure used includes the following stages:

1. Data Retrieval
   a. Download satellite imagery via *the* Copernican website. The downloaded image is Sentinel-2A from 2019 to 2023.
   b. NPLCT administration data of PT Arutmin Indonesia in the form of a shapefile (SHP).
2. Data Processing
   a. Cropping imagery to retrieve the area that the observation was conducted. Image cropping aims to simplify the analysis process.
   b. Radiometric correction is carried out to remove noise contained in the image as a result of distortion by the position of sunlight and clouds at the research site.

3. Data Analysis
   a. Changes in the greenish index in the observation area using the *Normalized Difference Vegetation Index* (NDVI) method which is based on a combination of pixel values of infrared band images or NIR and red bands. This method produces 7 classes of greenish level changes, namely land is not vegetated, low vegetation, rather low vegetation, medium vegetation, high enough vegetation, high vegetation and very high vegetation. In these 7 classes of greenness level change, there will be areas that experience reforestation, deforestation and do not change (Bashit, 2019).
   b. This greenness change refers to the Circular of the Chief Inspector of Mines (KAIT) No.9E/MB.07/DBT.PL/2023 concerning the spatial database of planning, implementation, and reporting of reclamation and post-mining in mineral and coal mining business activities.

Table 1. *NDVI value*

Class	Color	NDVI	Information
1	Blue	-1 to -0.1	Non-vegetated
2	Red	0.1 to 0.3	Low vegetation
3	Oren	0.3 to 0.5	Vegetation is rather low
4	Yellow	0.5 to 0.6	Medium vegetation
5	Light Green	0.6 to 0.7	Vegetation is quite high
6	Toska Green	0.7 to 0.9	High Vegetation
7	Dark Green	0.9 to 1	Very High Vegetation

This analysis is to take advantage of the combination of calculating the pixel value of infrared band or NIR and red band images from the visible header frequency. The scoring result or value is further divided into 3, namely reforestation, deforestation and no change (Cahyono, Nugroho, & Arifilla, 2019).

The division of these 3 scores is based on changes from 7 NDVI classes. Reforestation sees a change from unvegetated to very tall vegetation. The opposite of reforestation is deforestation which from very high vegetation levels to non-vegetated. If the rate of life does not change, it does not change. Results of NDVI interpolation from year i to year n to see what has not changed, deforestation and reforestation.

RESULTS AND DISCUSSIOn

Changes in the Greenness of the Land

The results of changes in the greenness level in PT Arutmin Indonesia's NPLCT area are as follows:

1. Sentinel-2A Image Processing

Sentinel-2A image processing obtained through the Copernicus website with a recording date of December 19, 2019 for the first image. The second image is recorded September 24, 2020. The third image recording date is June 21, 2021. The fourth image of the recording date is August 15, 2022. The fifth image of the recording date occurs October 29, 2023. The five images were cut according to the administrative area or location of the study. This cut was made to focus more on the research site (Fahreza, Aulia, Fauzan, Somantri, & Ridwana, 2022).

Figure 2. Research Location Benchmarking

After cutting the image, radiometric correction is performed. This radiometric correction is to reduce noise such as lighting, color and clouds contained in the location imagery. All five images are 100% cloud-clean, so no cloud radiometric correction is required on the imagery. Satellite resources are said to be good if the area of cloud coverage is less than 10% (Danoedoro et al., 2019).

Sentinel-2A Image Analysis

The *Normalized Difference Value Index* (NDVI) analysis method calculates the pixel value by combining the infrared band or NIR and the red band. The band combination continued to divide the greenish level into 7 classes, namely unvegetated land, very low vegetation, low vegetation, medium vegetation and high vegetation (Faizal et al., 2023). After the division of 7 classes of greenish level, the results of greenness changes were obtained by interpolation from 2019 to 2023. The results of this interpolation get 3 changes, namely reforestation, deforestation and no change.

Table 2. Greenness Level 2019 to 2023

Information	2019	2020	2021	2022	2023
Non-vegetated	15,35	14,52	12,34	14,08	15,91
Low vegetation	12,81	7,17	4,27	8,00	27,29
Vegetation is rather low	67,30	43,64	7,20	55,19	55,28
Medium vegetation	2,95	31,31	6,07	20,99	0
Vegetation High enough	0	1,76	14,47	0,15	0
High Vegetation	0	0	54,09	0	0
Very High Vegetation	0	0	0	0	0
Total	98	98	98	98	98

Greenish level results for sentinel-2A imagery in PT Arutmin Indonesia's NPLCT RPT area with an area of 98 ha in 2019 in the non-vegetated class covering an area of 15.35 ha, low vegetation covering an area of 12.81 ha, rather low vegetation 67.30 ha, medium vegetation covering an area of 2.95 ha, high enough vegetation, high vegetation and very high vegetation covering an area of 0 ha. Sentinel-2A image in 2020 for the unvegetated class covering an area of 14.52 ha, low vegetation covering an area of 7.17 ha, rather low vegetation covering an area of 43.64 ha, medium vegetation covering an area of 31.31 ha, moderately high vegetation of 1.76 ha, high vegetation and very high vegetation covering an area of 0 ha. The 2021 sentinel-2A image for the non-vegetated class covers an area of 12.34 ha, low vegetation covers an area of 4.27 ha, rather low vegetation covers an area of 7.20 ha, medium vegetation covers an area of 6.07 ha, moderately high vegetation covers an area of 14.47 ha, high vegetation is 54.09 ha and very high vegetation is 0 ha. The 2022 sentinel-2A image for the unvegetated class is 14.08 ha, low vegetation covering an area of 8.00 ha, rather low vegetation covering an area of 55.19 ha, medium vegetation covering an area of 20.99 ha, moderately high vegetation 0.15 ha, high vegetation and very high vegetation covering an area of 0 ha. Sentinel-2A 2023 imagery for the non-vegetated class covering an area of 15.91 ha, low vegetation covering an

area of 27.29, rather low vegetation covering an area of 55.28 ha, medium vegetation, moderately high vegetation, high vegetation and very high vegetation covering an area of 0 ha (Hasan, Widodo, Mutaqin, Taufik, & Hidayat, 2021).

Table 3. Greenish Rate Interpolation From 2019 to 2023

Information	Year				
	2019-2020	2020-2021	2021-2022	2022-2023	2019-2023
Reforestation	38,41	79,09	1,08	0,47	2,27
Deforestation	1,65	1,35	79,48	40,65	20,35
Remain	58,14	17,76	17,68	57,17	75,64
Total	98	98	98	98	98

The results of *the NDVI* image overlay from 2019 to 2020 obtained an area of land that experienced reforestation or an increase in greenness level of 38.41 ha. Deforestation or decrease in greenness levels that occurred covering an area of 1.65 ha. The area that has not changed is 58.14 ha. Seen in figure 3.

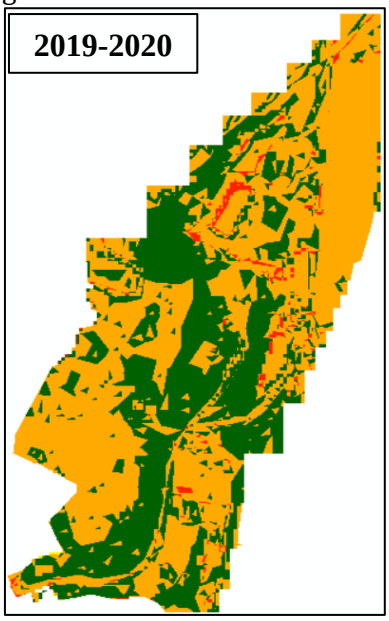

Figure 3. Greenish Level Interpolation from 2019 to 2020

The results of *the NDVI* image overlay from 2020 to 2021 obtained an area of land that experienced reforestation or an increase in greenness level of 79.09 ha. Deforestation or decrease in greenness levels that occurred covering an area of 1.35 ha. The area that has not changed is 17.76 ha. Seen in figure 4.

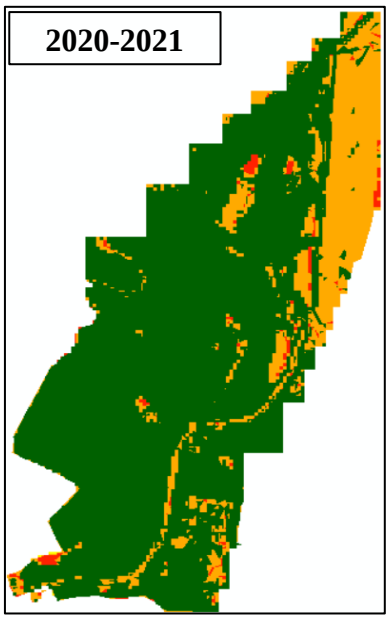

Figure 4. Greenish Rate Interpolation from 2020 to 2021

The results of *the NDVI* image overlay from 2021 to 2022 obtained an area of land that experienced reforestation or an increase in greenness level of 1.08 ha. Deforestation or decrease in greenness levels that occurred covering an area of 79.48 ha. The area that has not changed is 17.68 ha. Seen in figure 5.

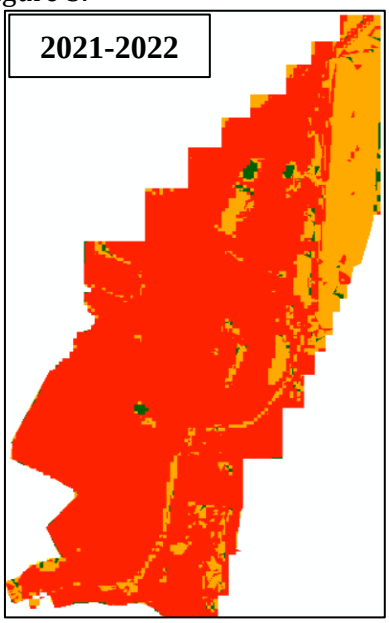

Figure 5. Greenish Rate Interpolation from 2021 to 2022

The results of the *NDVI* image overlay from 2022 to 2023 get an area of land that has undergone reforestation or an increase in greenness level of 0.47 ha. Deforestation or decrease in greenness levels that occurred covering an area of 40.65 ha. The area that has not changed is 57.17 ha. Seen in figure 6.

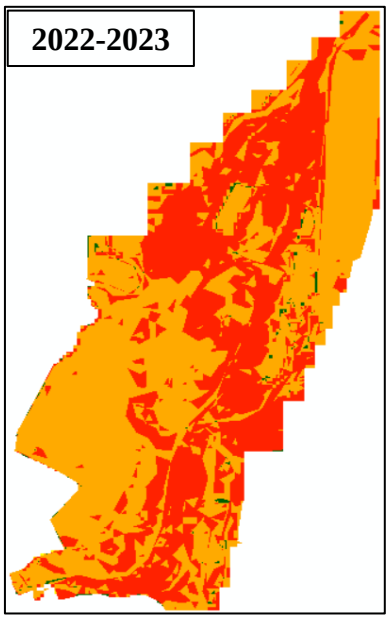

Figure 6. Greenish Rate Interpolation from 2022 to 2023

The results of *the NDVI* image overlay from 2019 to 2023 get an area of land that has undergone reforestation or an increase in greenness level of 2.27 ha. Deforestation or decrease in greenness levels that occurred covering an area of 20.35 ha. The area that has not changed is 75.64 ha. Seen in figure 7.

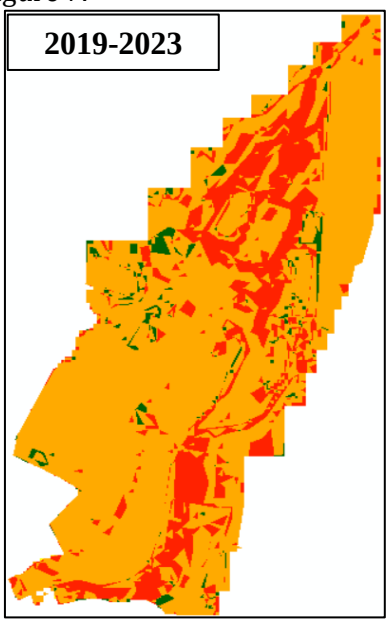

Figure 7. Greenish Rate Interpolation from 2019 to 2023

The graph of change shows that the largest area of 79.09 ha that experienced an increase in greenness occurred between 2020 and 2021. The rapid change in deforestation or decrease in greenness occurred between 2021 and 2022 with an area of 79.48 ha.

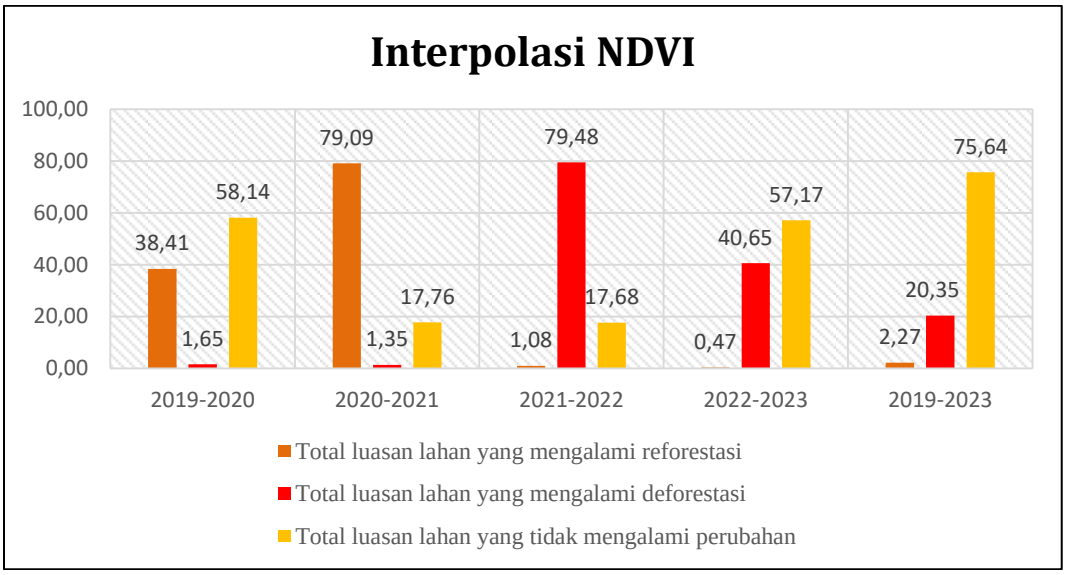

Figure 8. Interpolation graph of *NDVI changes* from 2019 to 2023

CONCLUSION

The greenness level in PT Arutmin Indonesia's NPLCT RPT area with sentinel-2A imagery in 2019 is the smallest area in the vegetation class is quite high, high vegetation and very high vegetation with an area of 0 ha, while the largest area in the vegetation class is rather low with an area of 67.30 ha. The sentinel-2A image in 2020 is the smallest area in the high vegetation class and very high vegetation with an area of 0 ha, while the largest area is in the rather low vegetation class with an area of 43.64. The sentinel-2A image in 2021 is the smallest in the very high vegetation class with an area of 0 ha, while the largest area is in the high vegetation class with an area of 54.09 ha. The 2022 sentinel-2A image is the smallest in the high vegetation class and very high vegetation with an area of 0 ha, while the largest area in the rather low vegetation class with an area of 55.19 ha. The 2023 sentinel-2A image is the smallest in the medium vegetation class, the vegetation is quite high, the vegetation is high and the vegetation is very high with an area of 0 ha, while the largest area is in the rather low vegetation class with an area of 55.28 ha.

Changes in land greenness in PT Arutmin Indonesia's RPT NPLCT area with sentinel-2A imagery from 2019 to 2020, 38.41 ha experienced reforestation, 1.65 ha experienced deforestation and 58.14 ha did not change. Sentinel-2A image from 2020 to 2021, 79.09 ha experienced reforestation, 1.35 ha experienced deforestation and 17.68 ha did not change. Sentinel-2A image from 2021 to 2022, 1.08 ha experienced reforestation, 79.48 ha experienced deforestation and 17.68 ha did not change. Sentinel-2A imagery from 2022 to 2023, 0.47 ha experienced reforestation, 40.65 ha experienced deforestation and 57.17 ha did not change. The sentinel-2A image changes from 2019 to 2023, 2.27 ha is reforested, 20.35 ha is deforested and 75.64 ha is unchanged.

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